Topic 2.7 - Composition of Functions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: ____________ Class: ____________ Date: ____________

Why this topic matters

A composition represents a process applied to the output of another process. The inside function acts first. A complete solution tracks the intermediate value, verifies that it belongs to the outer function's domain, and interprets the order in context.

Learning targets

- Evaluate compositions from formulas, tables, graphs, and contexts.
- Determine composition domains and explain why order matters.
- Construct and decompose composite functions while interpreting units and intermediate quantities.

Language and structure

Term	Meaning in this topic
composition	The function $(f \circ g)(x) = f(g(x))$.
inner function	The function evaluated first in a composition.
outer function	The function applied to the inner output.
composition domain	Inputs allowed by the inner function whose outputs are also allowed by the outer function.
decomposition	Expressing a function as a composition of simpler functions.

Launch: make a claim before calculating

Launch investigation

Reasoning first

A sensor converts a temperature t into a voltage $V(t) = 0.04t + 1.2$. A display converts voltage v into a reading $R(v) = 25v - 30$. Which composition represents the displayed reading from temperature, and what does the intermediate value mean?

Concept 1: Evaluate in the correct order

Core idea

In $(f \circ g)(x)$, compute $g(x)$ first and then apply f.

- Use parentheses when substituting the entire inner expression.
- For tables or graphs, locate the inner output and use it as the input to the outer function.
- $(f \circ g)(x)$ and $(g \circ f)(x)$ are generally different.

Worked example - annotate the reasoning

Problem. Let $f(x) = 2x^2 - 3$ and $g(x) = x - 4$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$.
Model reasoning. $(f \circ g)(x) = 2(x-4)^2 - 3$. $(g \circ f)(x) = (2x^2 - 3) - 4 = 2x^2 - 7$. They are not equal.

Check your understanding

Independent

If $f(5) = 12$ and $g(2) = 5$, find $(f \circ g)(2)$.

Concept 2: Determine the domain of a composition

Core idea

An input must be in the domain of the inner function and the inner output must be in the domain of the outer function.

- Square roots require nonnegative radicands.
- Denominators cannot equal zero.
- Logarithms require positive arguments.

Worked example - annotate the reasoning

Problem. Let $f(u) = \sqrt{u-1}$ and $g(x) = 5 - x^2$. Find $(f \circ g)(x)$ and its domain.
Model reasoning. $(f \circ g)(x) = \sqrt{4 - x^2}$. Require $4 - x^2 \geq 0$, so the domain is $[-2, 2]$.

Check your understanding

Independent

Let $p(u) = 1/(u+3)$ and $q(x) = 2x - 1$. Find the domain of $p(q(x))$.

Concept 3: Build and interpret a composite process

Core idea

Composition is useful when one output becomes the input to another rule.

- Name the intermediate quantity and its units.
- Order reflects the actual sequence of operations.
- Decomposition is not unique, but a useful decomposition should reveal the process.

Worked example - annotate the reasoning

Problem. A store applies a 15% discount and then a 5% service charge. Let $D(x) = 0.85x$ and $S(x) = 1.05x$. Write the final cost and compare it with applying the service charge first.
Model reasoning. $S(D(x)) = 1.05(0.85x) = 0.8925x$. Because multiplication commutes, $D(S(x))$ is also $0.8925x$ in this special case, though composition is not generally commutative.

Check your understanding

Independent

Decompose $H(x) = \sqrt{3x^2 + 7}$ into two simple functions.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1

No calculator

For $f(x) = x^2 + 1$ and $g(x) = 3x$, find $(f \circ g)(2)$.

Exit ticket 2

No calculator

Find the domain of $\sqrt{7 - 2x}$.

Exit ticket 3

No calculator

Why can $f \circ g$ exist at an input where $g \circ f$ does not?